

Boon Thu

AUTOMATION & ROBOTICS ENGINEER

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PROFESSIONAL SUMMARY

Automation Engineer focused on developing intuitive HMI interfaces, PLC (Siemens TIA Portal) logic, and AutoCAD instrumentation layouts. Proficient in DCS (Valmet DNA) and industrial robot simulation to drive operational efficiency and safety.

Portfolio

WORK EXPERIENCE

Tutor Assistant | 2026 Jan- Present

Institute of New Industry of Jyväskylä University of Applied Sciences

- Authored a comprehensive TIA Portal V19 (WinCC Unified) user guide detailing advanced HMI configurations, including UDT-based faceplates, security administration, parameter sets, and dynamic screen navigation.
- Guided 30+ students through TIA Portal and HMI design principles, actively debugging PLC logic (Ladder/FBD) and tag integration to ensure seamless hardware-interface communication.

EDUCATION

Bachelor of Engineering in Automation and Robotics | 2023-2027

Jyväskylä University of Applied Sciences

- Relevant Coursework: Programmable Logic Controllers (PLC), DCS (Valmet DNA), Linux, HMI, Control Systems, Machine Vision, Industrial Instrumentation, Electronics, Data Networks, Industrial Robots, Microsoft office Tools, R&D.
- GPA: **4.67 / 5** Credits: **193**

SKILLSET

Automation & Control

- PLC: Siemens TIA Portal (Ladder, FBD), Cascade Control strategies.
- DCS: Valmet DNA (Water process control).
- Safety: Risk and hazard calculation using SISTEMA.

Engineering Design & Visualization

- HMI/SCADA: Advanced GUI design in WinCC.
- CAD: AutoCAD (2D/3D layouts).
- UI/UX: Figma (Interface mockups and prototyping).
- Simulation: Visual Components (Robotic workflows, AMRs, Cobots integration).

Programming & Data

- Languages: C# (Modbus TCP/IP integration), MATLAB (Modeling/Scripting).
- Instrumentation: Sensor measurement and data acquisition via LabVIEW.

CERTIFICATE

- AI: Google AI Essentials Certified.
- IT Support: Google IT Support Certified (Troubleshooting, Maintenance)

EXCHANGE PROGRAM

E³UDRES² Bootcamp 2025 – St. Pölten, Austria

Theme: Creative Roots- Designing Experiences Inspired by Nature

- Co-developed a nature-inspired solution for regional climate and biodiversity challenges in collaboration with Natur im Garten
- Applied design thinking, rapid prototyping, and creative problem-solving in an international, multidisciplinary team
- Gained hands-on experience in sustainable innovation, intercultural collaboration, and human-centered design

VOLUNTEER EXPERIENCE

AgriVenture BioCamp 2025 – Volunteer Junior Staff

Jyväskylä University of Applied Sciences

- Supported event logistics by preparing the venue, setting up exhibition stands, and managing decorations
- Advised and guided international guests, contributing to smooth event coordination and a positive visitor experience

LANGUAGE SKILLS

Burmese (Native)

English (C1)

Finnish (A2)

Japanese (N4)
